Team Name : **SuryaSetu**

Team Leader Name : **Harsh Kawatra**

Problem Statement : **PS-15 | Nowcasting and Forecasting of Solar Flares from Aditya-L1 X-ray data**

SoLEXS 2-22 keV **HEL1OS 8-150 keV** **LightGBM + MiniROCKET** **TSS-driven** **Zero cost on a laptop**

Team Members

Team Leader
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Team Member-1
Name: Gursimran Kaur
College: Guru Gobind Singh Indraprastha University (GGSIPU)

A two-person, two-institute team: **full-stack and AI/ML engineering** *applied to ISRO's own solar mission.*

Opportunity and USP: why SuryaSetu wins

Most flare tools use a single channel from Earth orbit, and most blur detection and forecasting into one. **SuryaSetu is built differently on three axes:**

1. Dual-payload fusion.

SoLEXS measures **soft X-rays** (the heat a flare gives off) and HEL1OS measures **hard X-rays** (the instant energy is released).

We merge both onto a **single timeline** from the L1 vantage, so the model reads the **full physics** of a flare instead of half of it - *the reason we expect to outperform single-channel tools, which we prove with a single-vs-dual ablation.*

2. Physics-grounded lead time.

Hard X-rays surge several minutes before the damaging **soft-X-ray peak**, a real published relationship called the **Neupert effect**.

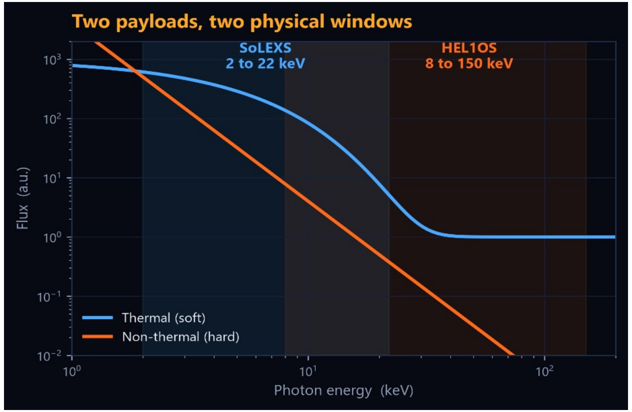
We convert that natural head-start into the actual **warning time** an operator needs, so the forecast rests on solar physics, *not a curve fit that only matches past data.*

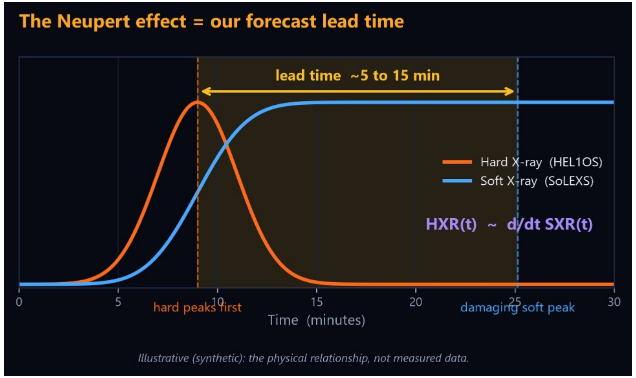
3. Honest, operational evaluation.

Flares are rare, so a model that always says "no flare" looks **99% accurate** and is useless.

We score with the **True Skill Statistic**, calibrate every probability, and validate only on **future time blocks** - the standard ISRO and the space-weather community actually trust.

We forecast the damaging peak - defensible physics - and stay honest that *true pre-onset prediction is the harder frontier*.





USP: the only dual-payload, L1-vantage system that turns the Neupert effect into calibrated, TSS-validated lead-time warnings, reproducible at zero cost on a laptop.

Features offered by the solution

01

Automated nowcast catalogue

Real-time soft and hard detection merged into one master catalogue: onset, peak, end, class and SNR.

02

Calibrated forecast + lead time

P(flare within N minutes) with calibrated confidence and a measured lead-time distribution.

03

Live dashboard + visual alerts

Dual light curves, hard/soft ratio and a green, amber, red banner that flips before the peak.

04

Full A-to-X dynamic range

Adaptive baselining plus dual SoLEXS detectors catch faint A/B/C and giant M/X flares.

05

Explainable alerts (SHAP)

Every forecast states which precursor fired it, so scientists can trust and audit it.

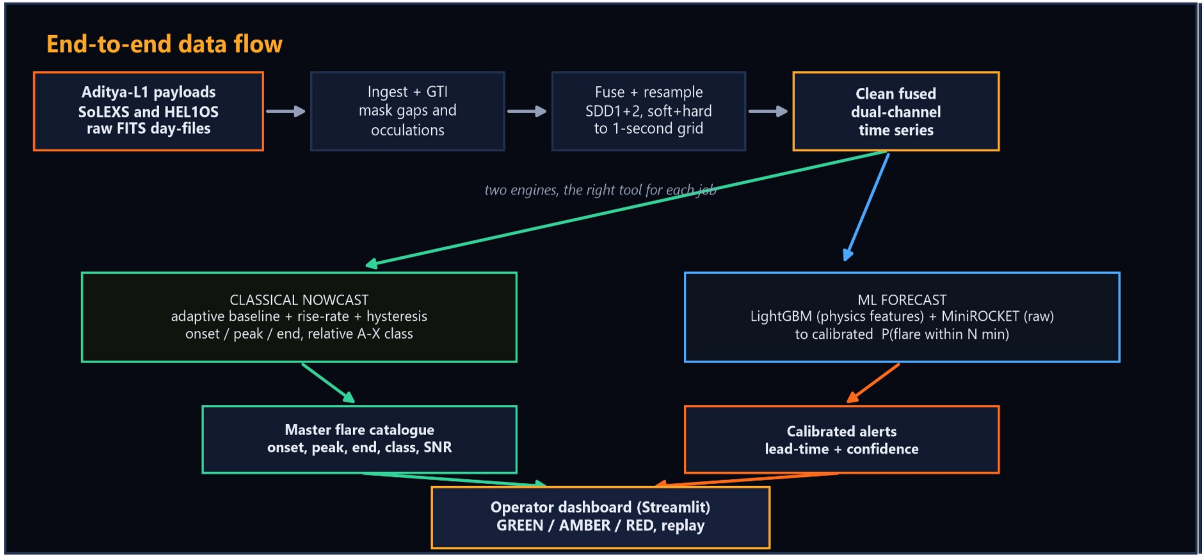
06

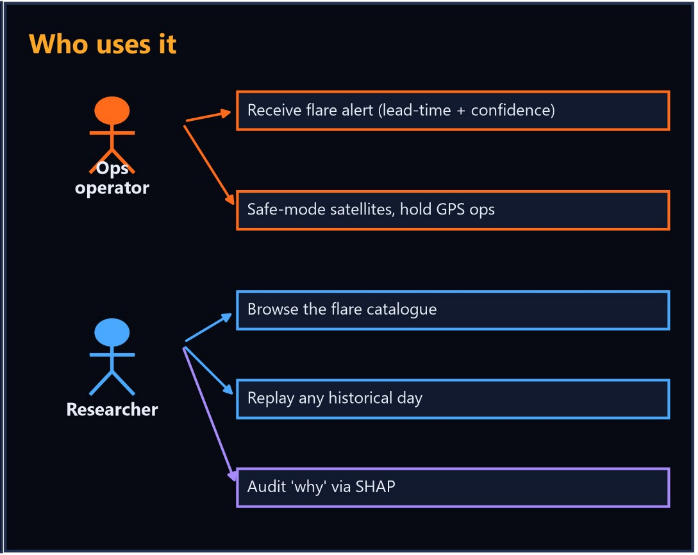
Offline and reproducible

No cloud, no cost, no proprietary software: end-to-end on one laptop from public data.

How it works in practice: the system reads both Aditya-L1 X-ray channels continuously, **automatically logs every flare** it detects into the catalogue, and **forecasts the next flare's strength and timing** with a **calibrated confidence**.
The operator watches one dashboard; when risk crosses a tuned threshold the **banner turns red** with the **minutes of lead time** remaining, and **SHAP** names the exact precursor that triggered it.

Process flow and use-case





In plain terms: raw satellite files are **cleaned** (bad seconds removed) and merged into one **1-second timeline**.

That stream splits into **two engines** running side by side: a **classical detector** that logs flares the moment they start, and an **ML forecaster** that predicts the next one.

Both feed the **catalogue, the alerts, and the live dashboard**.

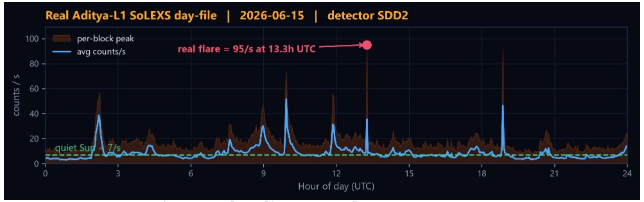
Two users.

An **operations engineer** receives an alert and safe-modes satellites or holds GPS-critical work;

a **researcher** browses the catalogue, replays any past day, and audits why each alert fired.

Wireframe: the operator dashboard

onset	peak	end	class	SNR	lead
06:12:04	06:19:31	06:41:10	M1.4	48.2	7.1m
05:48:22	05:52:09	06:03:55	C3.0	21.7	4.3m
04:31:50	04:38:12	05:02:40	C9.1	33.5	6.0m
02:09:14	02:13:40	02:31:08	B7.2	12.1	3.2m



Renders **real Aditya-L1 data** *(2026-06-15)*

One screen for the operator: the live soft and hard light curves with a **moving playhead**, the rising **hard/soft ratio** that acts as the early warning, a **probability gauge** for the next N minutes, and the running **flare catalogue**.

The banner **flips red** before the **soft-X-ray peak**, and *that gap is the warning time we deliver.*

QUIET

The Sun is **calm** and **no action is needed**.

The system keeps logging the **quiet-Sun baseline** it measures against.

WATCH

The **hard/soft ratio is climbing**, the physics precursor that often comes before a flare.

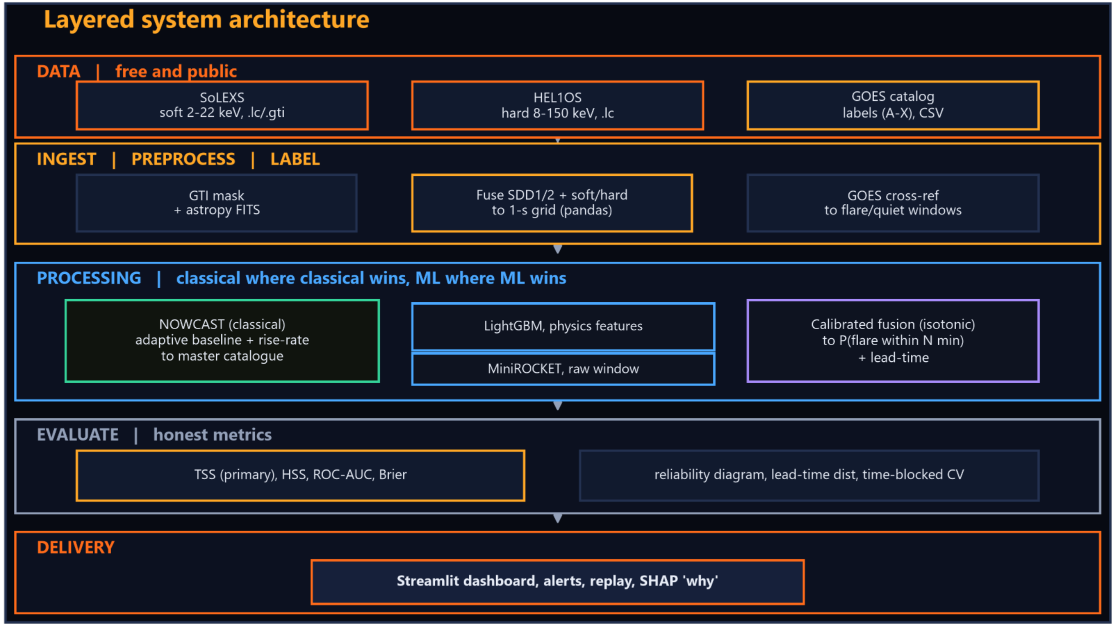
The operator is **put on notice** before anything is certain.

ALERT

A flare is forecast.

The banner shows the predicted **class**, the **minutes of lead time**, and a **calibrated confidence** so the operator can act.

Architecture of the proposed solution

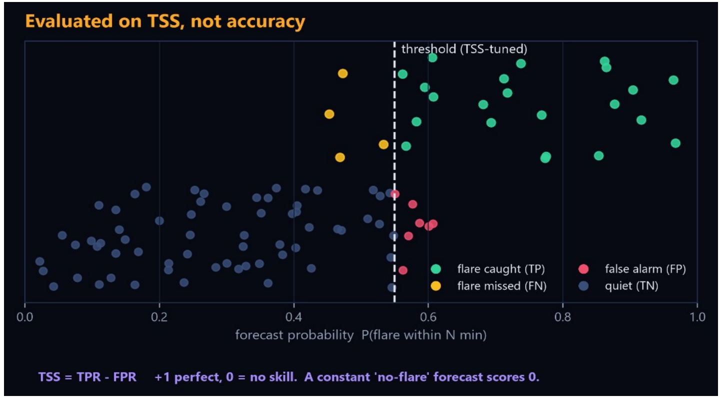


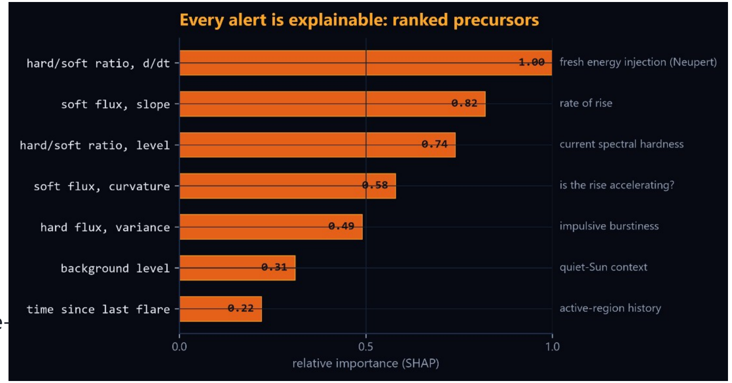
How it works: the cleaned soft and hard light curves feed **two paths at once**.

A **classical signal-processing detector** (adaptive baseline plus rise-rate) handles nowcasting, *interpretable and training-free*.

In parallel, a **LightGBM** gradient-boosted model on Neupert-physics features and a **MiniROCKET** convolutional time-series model on raw windows are fused and **isotonic-calibrated** into one probability, **P(flare within N minutes)**.

Every result is validated with **time-blocked cross-validation** on the **True Skill Statistic**, and **SHAP** attributes each alert to the precursor that drove it.





Both right-hand charts are illustrative - expected output, not measured results.

Technologies used in the solution

Layer	Tool	Why this tool	License
FITS I/O	`astropy`	standard reader for SoLEXS / HEL1OS products + GTI	BSD
Numerics	`numpy, pandas`	time-series alignment, resampling, fusion	BSD
Signal	`scipy`	detrend, smooth, derivative, adaptive baseline	BSD
Forecaster A	`LightGBM`	fast, interpretable, sample-efficient on rare events	MIT
Forecaster B	`MiniROCKET (sktime)`	CPU, deterministic, strong on raw time series	BSD
Calibrate, metrics	`scikit-learn`	isotonic calibration, TSS/HSS/ROC/Brier, blocked CV	BSD
Explainability	`SHAP`	per-alert precursor attribution (the why)	MIT
Figures	`matplotlib, plotly`	light curves, reliability, lead-time plots	BSD
Interface	`Streamlit`	fastest path to dual curves, alerts and replay	Apache
Environment	`Python 3.11 (uv)`	stable, prebuilt scientific wheels	PSF

Every layer, justified.

The prototype runs on a modest laptop (i5 / RTX 3050 / 16 GB) - proof that the method, not the hardware, does the work.

The same open-source stack scales straight onto ISRO's far larger infrastructure for full-mission throughput.

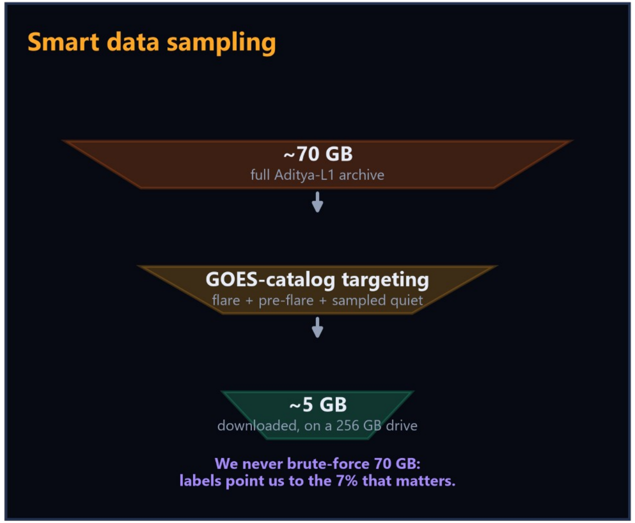
In ISRO's own tradition: the best results from the least resources.

Proof of work and open research: isro-hack-research.vercel.app [Click to Know] - a complete, first-principles companion to the science, data, and methodology behind SuryaSetu,
written and published so any engineer can understand and rebuild the system.

Estimated implementation cost

Prototype (our submission): effectively zero rupees. It needs only an internet connection and a modern laptop CPU (GPU optional), all of which we already own.

Item	Cost	Note
Aditya-L1 SoLEXS + HEL1OS data	Rs 0	free, ISSDC PRADAN (registered)
GOES flare catalog (labels)	Rs 0	free, NOAA SWPC
Software stack	Rs 0	all open-source (BSD/MIT/Apache)
Compute	Rs 0	existing laptop, i5 / RTX 3050 / 16 GB
Storage	Rs 0	existing 256 GB drive (~5 GB used)
Hosting (demo)	Rs 0	Streamlit local / free HF Spaces
TOTAL (prototype)	**Rs 0**	**only real input is engineering time**



Cost at ISRO scale *(operational, scoped on selection):*
compute at scale (GPU cluster or cloud) | **redundant full-mission storage** | **internet and data egress** | **engineering and MLOps staff**.

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THANK YOU